

Course Title: Introduction to Sustainable Chemical Engineering

Course Code: CHEXXX

Betul Bilgin
Alan Zdunek
Vikas Berry

CHE Faculty Retreat
July 31st, 2023



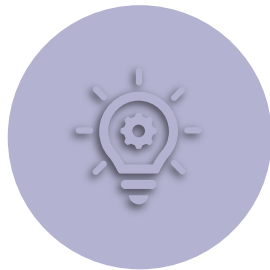
Course Description



Introduction to the principles and practices of sustainable chemical engineering.



Focus on five key themes: sustainable air, sustainable water, sustainable food, sustainable energy, and sustainable health.



Students will explore the challenges and solutions related to sustainability in these areas, applying chemical engineering principles and techniques.



The course aims to develop an understanding of the role of chemical engineering in addressing sustainability issues and promoting a sustainable future.

Course Objectives



1. UNDERSTAND
THE CONCEPT OF
SUSTAINABILITY
AND ITS RELEVANCE
TO CHEMICAL
ENGINEERING.



2. GAIN
KNOWLEDGE OF
THE
ENVIRONMENTAL,
SOCIAL, AND
ECONOMIC
CHALLENGES
RELATED TO
SUSTAINABLE AIR,
WATER, FOOD,
ENERGY, AND
HEALTH.



3. DEVELOP AN
UNDERSTANDING
OF THE ROLE OF
CHEMICAL
ENGINEERING IN
MITIGATING
ENVIRONMENTAL
IMPACTS AND
PROMOTING
SUSTAINABLE
PRACTICES IN THESE
AREAS.



4. ACQUIRE
PROBLEM-SOLVING
SKILLS AND
CRITICAL THINKING
ABILITIES SPECIFIC
TO SUSTAINABLE
CHEMICAL
ENGINEERING.

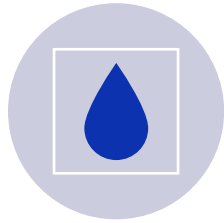


5. EXPLORE
INNOVATIVE
SOLUTIONS AND
TECHNOLOGIES FOR
SUSTAINABLE AIR,
WATER, FOOD,
ENERGY, AND
HEALTH.

Course Outline



SUSTAINABLE
AIR



SUSTAINABLE
WATER



SUSTAINABLE
FOOD



SUSTAINABLE
ENERGY



SUSTAINABLE
HEALTH

Each week will consist of lecture content, activity, discussion on movie/Youtube video/ article, Guest speaker, lab tour. Active learning, hand-on experiences and team-based learning will be implemented into course.

Week 1: Introduction to Sustainable Chemical Engineering

- Definition and importance of sustainability in chemical engineering
- Overview of the five sustainable themes: air, water, food, energy, and health
- Role of chemical engineering in addressing sustainability challenges
- Activity: What is Chemical Engineering? Who are chemical engineers? (write a story/ drawing)
- Potential Guest Speaker: UOP Renewable Energy

Week 2-3: Sustainable Air

- Air pollution and its impacts
- Chemical engineering approaches to air quality improvement
- Sustainable technologies for emission reduction
- Case studies on air pollution control in industries (Volkswagen emissions scandal)
- Activity: Design a air pollution sensor? Or design air filtration system?
- Potential Guest Speaker: EPA

Week 4-5: Sustainable Water

- Water scarcity and water pollution challenges
- Water treatment and purification processes
- Membrane technology for sustainable water management
- Case studies on sustainable water use in chemical processes (or Flint water crisis)
- Activity: Design a water purification system?
- Potential Guest Speaker: EcoLab/Nalco

Week 6-7: Sustainable Food

- Sustainable agriculture and food production systems
- Food processing and waste reduction strategies
- Packaging and preservation techniques for food sustainability
- Case studies on sustainable food supply chains
- Activity: Coffee Design by UC Davis
- Potential Guest Speaker: Tetra Pak

Week 8-9: Sustainable Energy

- Renewable energy sources and technologies
- Energy efficiency and conservation measures
- Integration of renewable energy in chemical processes
- Case studies on sustainable energy applications in chemical engineering
- Activity: Design a battery, design solar cell (Sanjay's design)
<https://academiccollegeprojects.com/renewable-energy-projects-for-engineering-students/>
- Potential Guest Speaker: Lanzatech, faculty U of Michigan
(<https://che.engin.umich.edu/research/research-areas/sustainable-energy/>)

Week 10-12: Sustainable Health

- Environmental health and chemical exposure
- Green chemistry in pharmaceutical and healthcare industries
- Sustainable waste management in healthcare facilities
- Case studies on sustainable practices for health and well-being
- Activity:
<https://users.rowan.edu/~hesketh/hesketh/cee%20drug%20delivery.pdf>
- Potential Guest Speaker: Abbvie

Week 13-15: Innovative Solutions for Sustainability

- Emerging technologies for sustainable air, water, food, energy, and health
- Life cycle assessment and sustainability evaluation methods
- Sustainable process design and optimization
- Technological advancements and future prospects
- Activity: Plant Tour
- Potential Guest Speaker: AIChE Sustainability Institute

Week 16: Review and Project Presentations



- Comprehensive review of course topics
- Final examination
- Project Presentation

Resources



Movies and TV shows:

1. Sustainable Air:

Movies:

- "Erin Brockovich" (2000) - Directed by Steven Soderbergh
- "A Civil Action" (1998) - Directed by Steven Zaillian

TV Shows:

- "The Newsroom" (2012-2014) - Created by Aaron Sorkin
- "Years of Living Dangerously" (2014-2016) - Executive produced by James Cameron

Reading list:

1. "Sustainable Air":

- "Air Pollution Control Engineering" by Noel de Nevers
- "Fundamentals of Air Pollution" by Daniel Vallero
- "Air Pollution: Health and Environmental Impacts" by Bhola R. Gurjar, Luisa T. Molina, and C.S.P. Ojha

YouTube Videos:

1. Sustainable Air:

- "The Problem with Air Pollution" by TED-Ed
- "Air Pollution 101" by National Geographic
- "How Does Air Pollution Affect Our Health?" by SciShow

Faculty Commitment, Tasks and Responsibilities

- For consistency and student engagement, it is recommended to have one faculty member teach the course. Multiple faculty members teaching may lead to reduced engagement and confusion among students, defeating the course's purpose.
- The faculty member teaching the course is responsible for content preparation, while other faculty members serve in an advisory role to design the contents for their corresponding modules.

Suggested Advisory Faculty List for each Module:

- Sustainable Air → Meenesh
- Sustainable Water → Brian
- Sustainable Food → Vivek
- Sustainable Energy → Sangil, Said, Anh
- Sustainable Health → Ying, Vikas
- Innovative Solutions for Sustainability --> Ezinne

Faculty Commitment, Tasks and Responsibilities

- During the module delivery, it is recommended that faculty members attend at least one lecture time, which may include guest speakers or engaging activities.
- Additionally, faculty members are encouraged to provide lab tours for the students.
- Throughout the semester, the course may feature social activities sessions, and it is suggested that faculty members participate in these activities.
- The overall commitment for faculty members includes 5-7 hours during the course design phase and 2-3 hours when the course is being taught during the semester.

Historical- CHE 101: “Introduction to Chemical Engineering Concepts”

Current Description and Topics

Current Catalog Course Description:

Overview of engineering and chemical principles used in chemical engineering technology. Thermodynamics, transport phenomena, and reaction engineering applied to process and product design. Course Information: Prerequisite(s): Credit or concurrent registration in [MATH 180](#); and Credit or concurrent registration in [CHEM 122](#) and Credit or concurrent registration in [CHEM 123](#). **3 credit hours:**

Course Learning Outcomes from Course Request System (CRS):

This course is intended to (1) present the historical origins and paradigms of chemical engineering, (2) prepare freshmen for the intellectual culture of chemical engineering through concrete problems and demonstrations that preview 200 and 300 level ChE requirements, (3) develop an understanding of the technological, societal, environmental and regulatory context of chemical engineering, (4) expose connections of chemical engineering to basic sciences and other engineering disciplines, and (5) indicate the role of chemical engineers in a variety of traditional industries and emerging fields.

Current CHE 101 Major Topics

- Genesis of chemical engineering as a field, paradigm of unit operations, historical achievements of chemical engineering, what chemical engineers do in varied industries- 3 hours
- Relation of ChE to sciences and other engineering fields, and technological frontiers in chemical engineering- 3 hours
- Units and conversions, dimensional analysis and dimensionless groups, conservation principle, example process flowsheet- 3 hours
- Shear stress and viscosity, mechanical work done by fluids, Reynolds number and flow regimes, history of thermodynamics, steam engines- 3 hours
- Phase behavior, vapor pressure curve, power of steam (demonstration), heat capacity of air (demonstration and calculation), distillation columns and operating lines- 3 hours
- Hydrogen generation and storage, hydrogen engines and fuel cells- 3 hours
- Diffusion (random walk – simulation), mass transfer coefficient, fluid mixing- 2 hours
- Thermodynamics of reactions, Gibbs energy and equilibrium constant, extent of reaction and simple equilibrium calculation- 3 hours
- Reaction kinetics and simple rate laws, CSTR and PFR reactors- 3 hours
- Environmental pollution, regulatory issues, hydrology and contaminant transport (including animated simulation)- 3 hours
- Membranes, desalination, wastewater treatment, nanotechnology concepts- 3 hours
- Heat transfer by conduction and convection, Newton's law of cooling (demonstration and calculation), wind chill factor- 3 hours
- Biochemical engineering concepts and applications- 3 hours
- Computers, programming and math packages: Tank drainage problem worked with Fortran, C, Java, Matlab, Maple- 3 hours

Quizzes & Exams- 4 hours

45 hours Total

CHE 101 Course Curriculum Proposal

- Change title to “Introduction to Sustainable Engineering”
- Change pre-requisites
- Revise course topics
- Add course to core CHE BS degree program